Worksheet 1: Lipid Digestion and Fair Comparisons

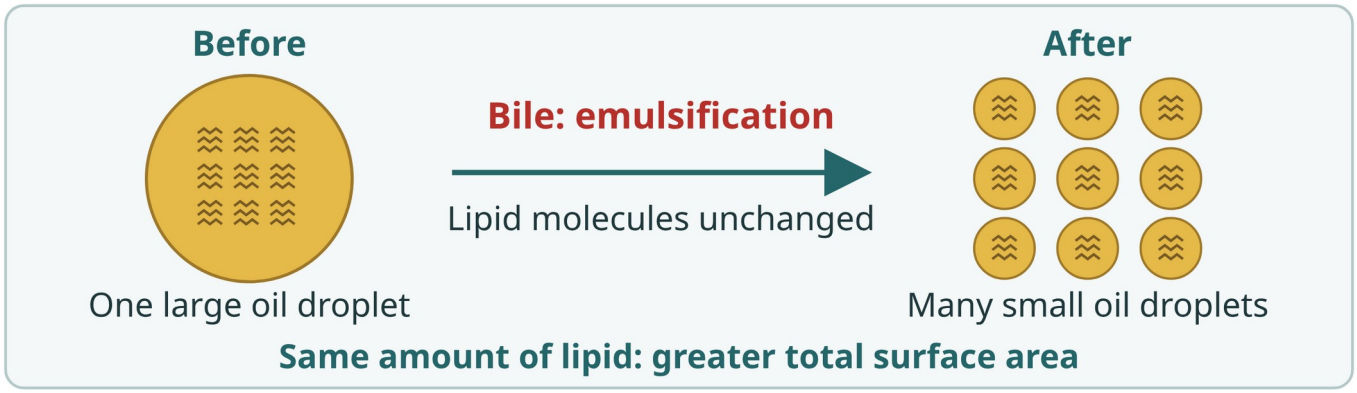
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Read the topic notes first, then apply the concepts to analyse the results.

Topic notes

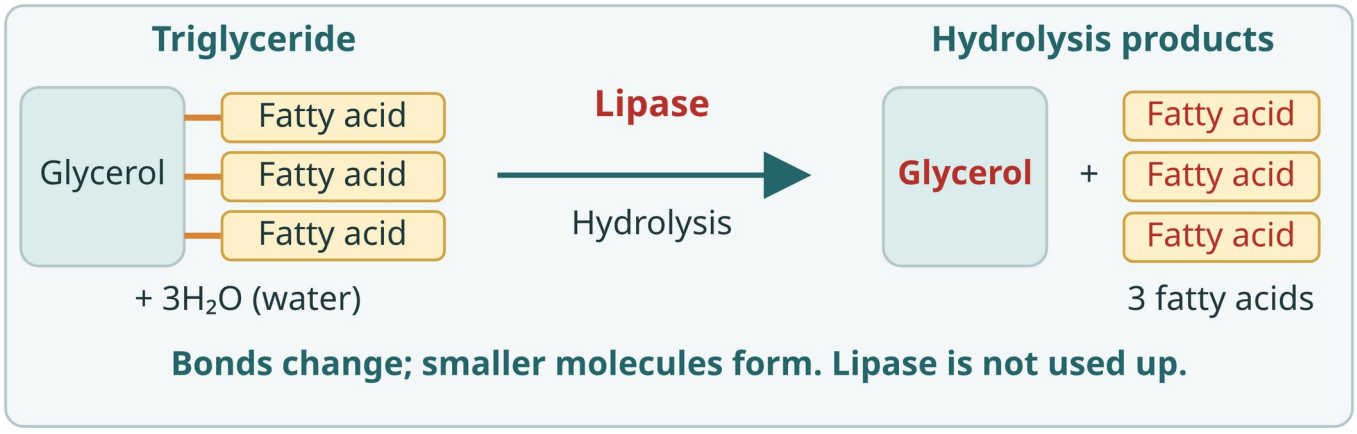
01 Bile: emulsifying lipids

Made in the liver and stored in the gall bladder, **bile** contains no digestive enzymes. It **emulsifies** large oil droplets into **small droplets**, increasing the **total surface area** for **lipase** to act on. This is **physical digestion**: lipid molecules remain unchanged.



02 Lipase in pancreatic juice: chemical digestion

Pancreatic juice containing **lipase**. **Lipase** catalyses **chemical digestion** of lipids. This course uses a simplified complete-hydrolysis model: **triglycerides** form smaller **glycerol** and **fatty-acid** molecules for absorption. **Lipase** is a **protein**; high heat can **denature** it and stop it working.



Context: lipids need digestion before absorption. What happens after adding digestive juice X or Y?



Part 1 | Context and fair comparisons

Researchers added 2 mL X or 2 mL Y to identical oil–water mixtures, shook them identically at 37°C, and compared appearances after one hour.

1. Identify the variables

Tick the appropriate factors in each row. You may choose more than one.

IV(the factor changed on purpose)；DV(the result you measure or observe)；CV(the factors kept the same)

Variable type	Factors to choose from
Independent variable Factor changed	☐ Total volume of oil and water ☐ Reaction temperature ☐ Reaction time ☐ Total volume of liquid added ☐ Method of shaking ☐ Appearance of the mixture ☑ **Combination of digestive juices**
Dependent variable Outcome measured	☐ Total volume of oil and water ☐ Reaction temperature ☐ Reaction time ☐ Total volume of liquid added ☐ Method of shaking ☑ **Appearance of the mixture** ☐ Combination of digestive juices
Controlled variables Kept the same	☑ **Total volume of oil and water** ☑ **Reaction temperature** ☑ **Reaction time** ☑ **Total volume of liquid added** ☑ **Method of shaking** ☐ Appearance of the mixture ☐ Combination of digestive juices

2. Select the assumptions for a fair comparison (choose all that apply)

(An assumption is a condition we have not proved but take as true, to help make the experiment results valid.)

☑ **All sets start with the same amounts of oil and water, and receive the same total volume of added liquid.**

☐ The X setup may receive more total liquid than the Y setup.

☑ **All sets are shaken in the same way.**

☑ **All sets are observed after the same reaction time.**

3. Design a control group

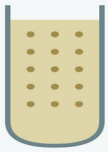
Does this experiment need a control group? If so, how would you design the comparison setup?

(Hint: A control group provides a starting point for comparison. It helps you check that changes are caused by the independent variable, rather than other outside factors. The control group therefore does not receive the treatment being tested.)

Use an identical oil–water mixture with 2 mL water as a baseline, compared with X or Y; keep initial amounts, volume, temperature, time and shaking equal.

Part 2.1 | Analyse the individual actions of X and Y

Model results one hour after shaking.

Set X	Set Y
Add 2 mL X	Add 2 mL Y
	
Cloudy emulsion	Thinner oil layer; two layers remain

4. What does digestive juice X do?

Digestive juice X mainly causes lipids to form a (**cloudy emulsion** / clear solution).

Part 2.2 | Analyse how X helps Y act

Two fresh tubes received 1 mL X or water and identical shaking, then 1 mL Y each. Results show one hour later.

X first, then Y	Water first, then Y
1 mL X → 1 mL Y	1 mL water → 1 mL Y
	
Less cloudy No obvious oil layer	Thinner oil layer Two layers remain

5. How does digestive juice X help Y act?

X disperses large oil droplets into small droplets. Their total surface area (**increases** / decreases), helping Y act.

6. What can this experiment tell us?

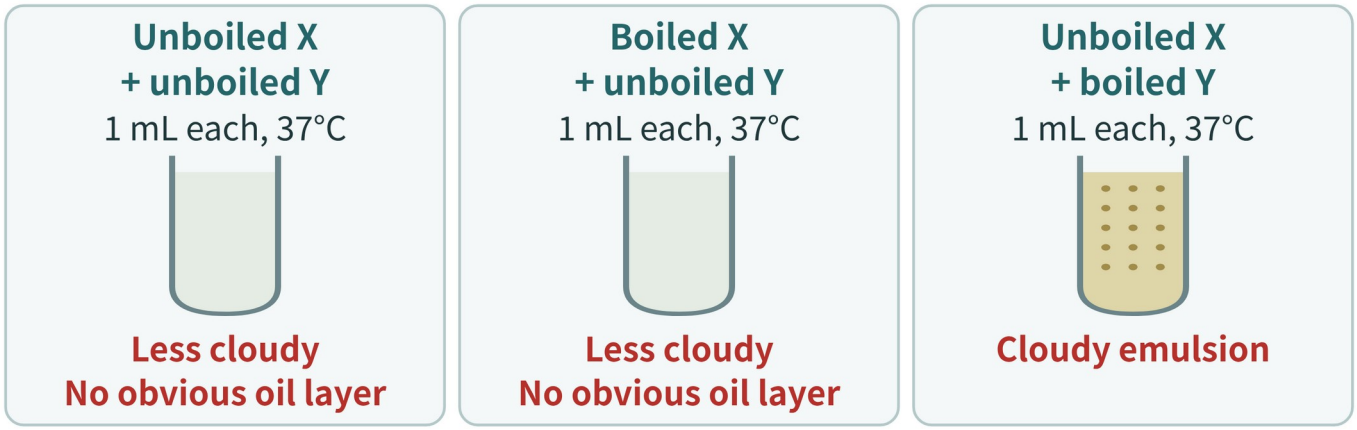
We observed only the oil layer, cloudiness or clarity, without testing digestion products. Which statement is most appropriate?

A. Appearance gives clues about the juices, but alone cannot prove that glycerol and fatty acids have formed.

B. A clear mixture fully proves that lipids have broken down into glycerol and fatty acids.

Part 2.3 | Effects of heating the juices

Use unboiled X then Y as the baseline. Boil one juice, cool to 37°C, then add 1 mL each of X and Y in the same order. Compare after one hour.



7. Extension conclusion

The juice that loses its action after boiling is (digestive juice X / **digestive juice Y**).

Part 3 | Integrated explanation

8. Using the experimental results and topic notes, explain how bile and lipase in pancreatic juice work together to help digest lipids, and how boiling affects their actions.

Bile physically emulsifies large droplets into small droplets without changing lipid molecules. The greater total surface area helps lipase act. Lipase in pancreatic juice catalyses chemical digestion into glycerol and fatty acids in this model. Lipase is a protein; boiling denatures it and stops its catalytic action even after cooling. Bile retains its emulsifying action after boiling.